Understanding the Dataset

HDI Prediction System | Epic 3: Dataset Download and Understanding

Objective

Explore the structure, quality, and distribution of the HDI dataset before deciding which columns to use as model features.

Code: Basic Exploration

```
df.shape          # (195, 880)
df.info()         # column types and non-null counts
df.describe()     # summary statistics for numeric columns
df.isnull().sum() # missing values per column
```

Key Observations

Column (2021 snapshot)	Missing Values	Notes
Country	0	Fully populated identifier
Human Development Groups	4	4 territories not classified by UNDP (e.g. small states)
HDI Rank (2021)	4	Same 4 unranked territories
Human Development Index (2021)	4	Target variable; 191 of 195 rows usable
Life Expectancy at Birth (2021)	0	Fully populated
Expected Years of Schooling (2021)	2	2 missing values
Mean Years of Schooling (2021)	4	4 missing values
Gross National Income Per Capita (2021)	2	2 missing values

Summary Statistics (2021 indicators used as model features)

Indicator	Min	Median	Max
HDI Rank (2021)	1	96	191
Gross National Income Per Capita (2021)	731.8	12,306.3	146,829.7

Data Quality Decisions

- Rows with a missing 2021 HDI value (the target) are dropped before training, since they cannot be used to supervise the regression model.
- The 4 core feature columns (life expectancy, expected years of schooling, mean years of schooling, GNI per capita) have very few missing values (0-4 out of 195), so rows with missing feature values are dropped rather than imputed, to avoid introducing bias.

- Only the most recent year (2021) is used for training the single-year prediction model; historical years (1990-2020) remain available in the dataset for potential future trend analysis but are not used by the current model.
- Columns unrelated to the four core features and the HDI target (e.g. Gender Inequality Index, CO2 emissions per capita) are excluded from the model input, since app.py only collects four values from the user.

Code: Missing Value Check for Core Columns

```
core_cols = [
    'Human Development Index (2021)',
    'Life Expectancy at Birth (2021)',
    'Expected Years of Schooling (2021)',
    'Mean Years of Schooling (2021)',
    'Gross National Income Per Capita (2021)'
]
df[core_cols].isnull().sum()
```

Next Step

With the dataset understood and cleaned of rows missing core values, proceed to **Selecting Dependent and Independent Variables** to prepare the feature matrix (X) and target vector (y) for model training.